



CEN429 Secure Programming

Syllabus — 2026-2027 Fall

Asst. Prof. Dr. Uğur CORUH

Course information

Contact	ugur.coruh@erdogan.edu.tr — subject line: [CEN429]
Office	F-301 · meetings by e-mail appointment
Class	Friday 09:00–12:00 · Faculty of Economics and Law Building, D-402
Website	ucoruh.github.io/cen429-secure-programming
Classroom code	A new code each term; announced in week 1
Language / type	Turkish · Elective · 7th semester
Hours / credits / ECTS	3 · 3 · 5
Prerequisite	CEN107 Algorithms and Programming I

Aim of the course

- **Recognise** common software vulnerabilities: buffer overflow, memory errors, injection
- **Apply** software protection methods: data security and cryptography, C/C++ and Java hardening, runtime self-protection, code obfuscation, whitebox cryptography
- Know security standards and **penetration test planning**
- Every topic: **faulty code** → **attack** → **fix**

Learning outcomes

Code	Learning outcome
LO.1	Identifies and classifies common vulnerabilities
LO.2	Explains encryption methods and the principles of secure communication (SSL/TLS)
LO.3	Applies code hardening techniques (input validation, safe memory, RASP, obfuscation) in C/C++ and Java
LO.4	Explains how to build a secure channel with encryption and authentication
LO.5	Produces a software plan based on secure design principles
LO.6	Knows security review and vulnerability assessment methods
LO.7	Knows secure programming standards and penetration test planning

Weekly schedule (1–8)

Wk	Date	Topic
1	18.09	Introduction to secure programming and the application protection plan
2	25.09	Computer viruses and security models
3	02.10	Data security: in transit, at rest, in use
4	09.10	Code hardening: C/C++
5	16.10	Code hardening: Java and interpreted languages
6	23.10	Runtime application self-protection (RASP)
7	30.10	Midterm project demonstrations
8	31.10–08.11	Midterm exam week — Quiz-1

Weekly schedule (9–16)

Wk	Date	Topic
9	13.11	Advanced obfuscation and diversification
10	20.11	Certificates and cryptographic methods
11	27.11	Whitebox cryptography
12	04.12	Security certifications and penetration test planning
13	11.12	Security requirements
14	18.12	Tigress and diversification
15	25.12	Final project demonstrations
16	04–17.01	Final exam period — Quiz-2

Assessment

- **One term project:** a C/C++ application + a security guide, with two checkpoints (rubric-based)
- **Two quizzes:** in the exam weeks

Component	Weight	When
Project checkpoint 1 (RAP1)	60% of the midterm	Week 7
Quiz-1	40% of the midterm	Week 8
Project checkpoint 2 (RAP2)	70% of the final	Week 15
Quiz-2	30% of the final	Week 16

Course grade = $0.4 \cdot \text{Midterm} + 0.6 \cdot \text{Final}$

Resources and tools

- **Lecture notes** (website) — the main source, sufficient on their own
- **Textbook:** J. Viega, M. Messier, *Secure Programming Cookbook for C and C++*, O'Reilly, 2003
- **Open standards:** SEI CERT, OWASP, MITRE CWE, NIST FIPS 140-3
- **A laptop is required:** WSL2 + Ubuntu, GCC/Clang, CMake, gdb, OpenSSL 3, SQLite, JDK 21, Tigress
- Demos: in the `code/week-NN` folders; build once inside `code`, then run `demo.ps1` (Windows) or `demo.sh` (WSL/Linux)

Rules

- **Late submissions are not accepted**; report unexpected circumstances immediately
- **Academic integrity**: working together is allowed; sharing solutions and uncited quoting are not
- **Communication**: university e-mail, with **[CEN429]** in the subject line
- **Ethics**: attack techniques are tried **only on your own computer** and only on the demos provided

Questions and answers

Course site: ucoruh.github.io/cen429-secure-programming